Some Anti-Patterns
(aka Don’t Do These!)

category	code	description
1	1	ordered
1	2	cancelled
1	3	shipped
2	Z	unspecified
2	A	Over 5 million
2	B	3 - 5 million
2	C	1 - 3 million
2	D	500,000 – 1 million
2	E	Less than 500,000
3	1	NorthEast Region
3	2	Western w/o CA
3	3	Southern & W.Indies
4	CP	Central private
4	CA	Central automated
4	CC	Climate controlled
4	CD	distribution

One True Lookup Table

- Common Lookup Table
- Allowed Value Table
- Massively Unified Code-Key (MUCK)

Idea: Combine all lookup tables into one

Supposed benefits:

- Fewer tables = simpler

Problems:

- Uniform data representation (attributes, types)
- Error-prone – easy to access wrong data.
- FKs must include category.
- Joins use larger lookup table.

Better design:

- Separate lookup tables

Entity Attribute Value / Object Attribute Model

emp_id	property	value
101	first_name	David
101	last_name	Bristol
101	date_of_birth	11/2/1946
102	first_name	Mary
102	last_name	Manning
102	date_of_birth	10/10/1951
103	first_name	Alistair
103	last_name	Ross
103	date_of_birth	5/21/1966

Idea: key, attribute_name, value

Supposed benefits:

- Simple to design
- Space-efficient for sparse data – no NULLs
- Can combine data from multiple tables

Problems:

- No constraints on data
- Uniform data representation
- Complex, inefficient queries

Better:

- Traditional design

Repeated attributes

fname	lname	home	work	cell	fax	pager
George	Barnes	562-874-1234		310-999-3628		
Susan	Noble	562-975-3388	714-847-3366			
Erwin	Star				714-997-5885	
Alice	Buck		562-577-1200	562-561-1921		
Frank	Borders	714-968-8201				
Hanna	Diedrich			562-786-7727		

Multi-value attributes

id	firstname	lastname	hobbies
1639	George	Barnes	reading
5629	Susan	Noble	hiking, movies
3388	Erwin	Star	hockey, skiing
5772	Alice	Buck	
1911	Frank	Borders	photography, travel, art
4848	Hanna	Diedrich	gourmet cooking

Attribute Splitting

sales_person	jan	feb	mar	apr
R.M. Pollock	2456	2285	1143	2159
R. Jones	2665	2998	3100	3187
J. Mehta	5781	4112	1954	2574
Anif Khan	2143	3124	2087	2694

OR

JanSales(sales_person, sales)

FebSales(sales_person, sales)

…

Supposed benefits:

- Corresponds to typical data presentation

Problems:

- Need to repeat constraints, queries
- Complicates queries
- Inflexible design
  - Hard-codes possible values
  - Leads to more NULLs

Better:

- Have explicit attribute in table